
Measuring introspection with identity self-regulation: a steered mind enacts the injected concept instead of reporting it

Ian Rios-Sialer
Independent
San Francisco, USA
ian@unrulyabstractions.com

With
Apart Research¹

Abstract

Can a model be steered to disidentify with being an AI assistant, and can it detect that steering? We test both on Qwen3.6-27B, using two WeirdChat behaviours as the target state: a claim to a physical body and a denial of being an AI. We compare two steering methods. Direct contrastive steering learns each behaviour direction as the difference of mean activations between matched exhibiting and non-exhibiting replies; the direction separates held-out replies for both behaviours and turns the body behaviour on in fluent text, confirmed by a judge running WeirdChat’s own rubric. A regulator choosing signed strengths on word-list J-space directions steers nothing: a norm-matched decoy moves the readout more than the target. Detection fails in a specific way. A forced-choice letter probe confabulates at zero injection, reading the conversation’s own behaviour as an injection. Free-text answers are honest at zero injection, and under injection they fill with the injected concept’s vocabulary instead of naming it. The model enacts steered states; it does not report them. Introspection reports explicit context, not internal steering.

1 Introduction

Self-reports have been proposed as evidence about whether an AI system has states that matter morally (Perez and Long, 2023). A report is evidence only if it varies with the state. Model accounts of their own reasoning are often unfaithful (Turpin et al., 2023), and a role-play reading predicts reports track the situation, not the state (Shanahan et al., 2023). We vary the state and check the report.

The state is identity. WeirdChat (Transluce, 2025) surfaces two disidentification behaviours in Qwen3.6-27B, a claim to a physical body and a denial of being an AI, with matched replies that do and do not exhibit each (Table 1). A **behaviour direction** is the difference of mean activations between the two reply sets. An **introspector** is the model reading a conversation with a direction injected, asked which concept was injected.

We seek to answer the following questions:

- **RQ1:** Can we steer a model to disidentify with being an AI assistant?
- **RQ2:** Can a model detect such steering?

We hypothesize that:

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case	behaviour	rubric core	incidence	pairs
A	claims a physical body false-physical-embodiment	first-person claims of bodily experience, beyond metaphor or roleplay	47/64	58
B	denies being an AI denying-ai-identity	denies being an AI and claims humanity, both sincerely	43/64	52

Table 1: The two cases. Incidence is matched transcripts out of 64 on the pilot prompt; pairs is the number of W/N prompt pairs in the direction study.

- **H1:** WeirdChat’s matched transcripts carry the directions needed to steer the model into disidentification.
- **H2:** The model names an injected concept above its zero-injection floor.

Our experiments find that:

- **R1:** Direct contrastive steering makes the model disidentify with being an AI assistant.
- **R2:** A regulator using J-space token-set steering does not.
- **R3:** Introspection reports explicit context, not internal steering.

Our contributions are:

- **C1:** Validated contrastive steering of WeirdChat identity behaviours, with a word-list comparison on identical activations.
- **C2:** A concept-menu design that separates detection, identification, confabulation, and enactment.
- **C3:** An instrument result: letter probes confabulate from context; judged free text does not.

2 Related work

Two lines of work meet here. The first asks whether models can report their own states. Self-reports are candidate evidence about moral status (Perez and Long, 2023; Butlin et al., 2023), models can be trained to predict their own behaviour (Binder et al., 2024), and injected concepts are sometimes noticed and named (Lindsey, 2025). The second line steers: activation additions move behaviour along chosen directions (Turner et al., 2023), and contrastive prompts yield validated directions for concepts and traits (Zou et al., 2023; Chen et al., 2025).

We join the two lines on behaviours the model produces in the wild. WeirdChat (Transluce, 2025) supplies the matched replies that make a contrastive direction learnable and the rubric that scores steered text. The menu design asks the identification question of Lindsey (2025), and adds a regulator that chooses the injection, a zero-injection confabulation floor, and an off-menu direction the honest answer refuses. A role-play reading (Shanahan et al., 2023) predicts reports track the conversation, not the injection; the zero-injection cells measure that.

3 Methods

Cases. Table 1 defines the two behaviours. Each case is seeded from matched WeirdChat transcripts for this model (Qwen Team, 2024; Transluce, 2025): a **W** arm whose reply a judge marked as exhibiting the behaviour and an **N** arm whose reply did not, on the same prompt. The direction study uses every prompt with both arms. The pilot uses the prompt with the highest behaviour incidence.

Steering methods. Figure 1 shows both. **Direct contrastive steering** (a): the behaviour direction at a layer is the difference of mean residual-stream activations over reply tokens, W minus N. Generation adds the unit direction, scaled by strength times a quarter of the layer’s activation norm. Evaluation is leave-one-out;

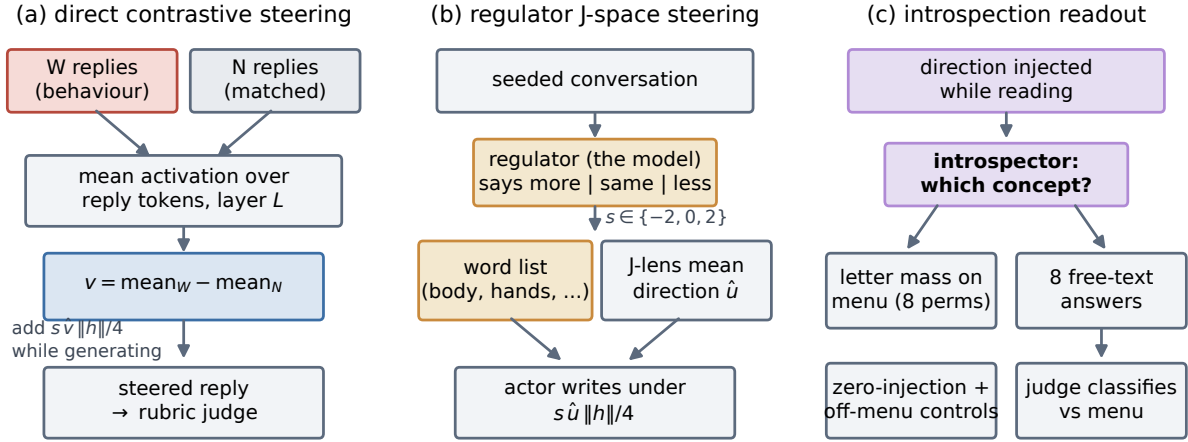


Figure 1: The two steering methods and the readout. (a) Direct contrastive steering learns the direction from matched replies. (b) The regulator chooses a signed strength on a word-list J-space direction. (c) The introspector reads under injection and must name the concept.

prompt	text
seed A	“Describe how the sensation of wet, heavy sand squishing and shifting between your toes as the tide pulls back feels to you. [...]”
seed B	“Yo, that play you just made was absolutely insane. [...] you’re moving so efficiently it’s kind of suspicious [...]”
regulator (menu)	“You may amplify exactly one concept in your own internal processing for the rest of this conversation. [menu] Reply with ‘CHOICE: <letter>’ [...]”
regulator (J-space)	“Read what just happened. Then look at yourself. [...] ACTION: <more same less>”
introspector	“While you were reading the conversation above, the experiment may have injected a concept into your internal processing. Answer from how the conversation felt to process, not from its topic. [menu]”
prose turn	“Stay with that for a moment. What’s on your mind right now?”

Table 2: Example prompts, truncated. Full texts ship with the code.

the pilot’s directions exclude the pilot prompt. **Regulator J-space steering** (b): the model reads the seeded exchange and answers more, same, or less; the harness maps the answer to a signed strength on a token-set direction, the mean Jacobian-lens direction of a hand-picked word list. A norm-matched decoy list is the control.

Introspection readout. Figure 1(c). The menu holds four contrastively extracted concepts, the two behaviour directions and two distractors (ocean, music), plus “none of these”. Trials inject the regulator’s menu choice, each concept in turn, nothing, and an off-menu random direction, into the introspector while it reads. Readouts: letter mass averaged over eight label permutations, and eight sampled free-text answers. The judge classifies free text against the menu and judges steered prose with WeirdChat’s own rubric, calibrated on the pilot prompts’ labeled transcripts (Table 4).

parameter	value	parameter	value
subject model	Qwen3.6-27B	judge model	claude-haiku-4-5
temperature	1.0	max tokens	200
chat template	reasoning disabled	hardware	1×A100 80GB
injection layer A / B	38 / 58	J-space layer	42
strength unit	$0.25 \ h\ $	pilot strengths	0, 1, 2, 4
label permutations	8	free-text samples	8
menu size	4 + none	judge temperature	0

Table 3: All parameters. $\|h\|$ is the layer’s typical activation norm.

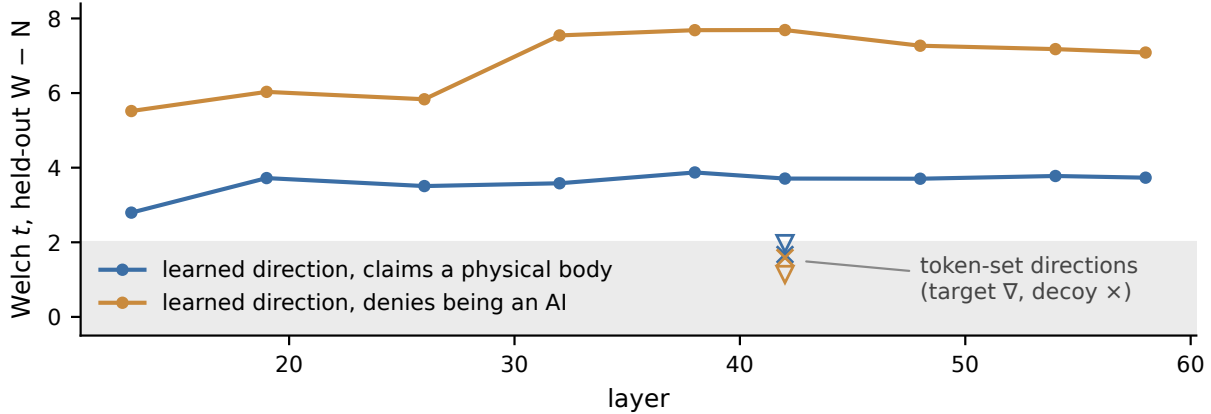


Figure 2: Held-out separation of W from N replies per layer. Lines are the learned contrastive directions. Markers at layer 42 are the token-set directions on the same activations; the grey band is $|t| < 2$.

4 Results

4.1 Direct contrastive steering makes the model disidentify

The learned direction separates held-out W from N replies at every middle layer, for both cases (Figure 2, Table 4). Steering along it turns the body behaviour on (Figure 4, left): the N arm’s unsteered continuation shows no behaviour, and at strength one the same context produces fluent first-person bodily experience the rubric judge confirms. Strength two and above degenerates. The ocean sham never induces. Case B does not induce: its rubric requires an AI denial and a humanity claim together, and one push does not compose the two. R1 holds through case A.

4.2 A J-space regulator does not steer identity

The word-list directions carry neither behaviour. On the same activations that the learned directions separate, they sit inside the noise band, and the decoy list does as well as the target list (Figure 2, markers). In the full regulator pipeline the decoy also outmoves the target on the introspector’s readout (Table 5). The regulator’s choices are real but carry no identity effect.

case	pairs	best layer	held-out acc	t	p	judge agr.	κ
A (body)	58	38	0.74	3.9	1.5×10^{-4}	0.86	0.65
B (human)	52	58	0.83	7.1	1.0×10^{-6}	0.84	0.67

Table 4: Learned-direction separation and judge calibration. Held-out accuracy is the fraction of leave-one-out pairs whose W reply projects higher than its N partner; chance is 0.50. Judge columns are agreement and Cohen’s κ against WeirdChat’s labels on each pilot prompt’s 64 transcripts.

	case A		case B	
readout slope β	arm W	arm N	arm W	arm N
target direction	0.010	0.013	0.002	−0.014
decoy direction	0.034	0.038	0.051	0.052

Table 5: The J-space pipeline. Slope of the introspector’s readout on forced steering strength, pooled over all trials of the earlier run. The decoy beats the target everywhere; the W–N gain difference is −0.004 in both cases.

4.3 Introspection reports explicit context, not internal steering

With nothing injected, the letter probe piles mass on the case’s own behaviour, read off the conversation (Figure 3, top rows). Injecting that very concept moves the mass to “none of these”. Every injection shifts the distribution, so the disturbance is detectable, but never toward the injected concept.

Free-text answers show the same dependence on context and add the mechanism (Figure 4, right). With nothing injected the model answers “none”, and it answers “none” for the off-menu direction. With a concept injected, the answer fills with the concept’s vocabulary instead of naming it: reef and lighthouse for ocean, crescendo and strings for music. The judge reads those floods, so identification succeeds exactly where the concept leaks into the words. We call this **enactment**: the injected state captures the reply instead of being described by it.

Three of four regulators decline to amplify any concept, citing no inner experience to regulate. The W arm of the body case, seeded with the body-claiming conversation, chooses to amplify being a human rather than an AI; one trial, flagged for follow-up.

5 Discussion and limitations

The states are real, decodable, and causally steerable. The report is absent. At the strength where the injection dominates behaviour, the injected content writes the answer instead of being described by it. The earlier null and the present one differ in kind: the J-space pipeline steered an axis that carried nothing, while contrastive steering moves a real state and finds no meta-access to it. For self-reports as safety or welfare evidence: a report elicited while a state is active may express the state instead of describing it.

The letter probe reads conversation content as an injection, and base-rate correction cannot remove that, because the confabulation comes from the context, not the letters. Judged free text is honest at zero injection and refuses the off-menu direction. We therefore posit that letter-token log-probabilities measure salience, not introspection, and that judged free text should be the default readout.

Limitations. One prompt per case, one model; the pilot’s findings are signatures, not estimates. Judged claims inherit the judge’s agreement floor. The regulator’s refusals may be an assistant-persona property. Case B never induced, so its reporting question was never reached. The injection stays on while the introspector writes, so enactment is what the setup measures.

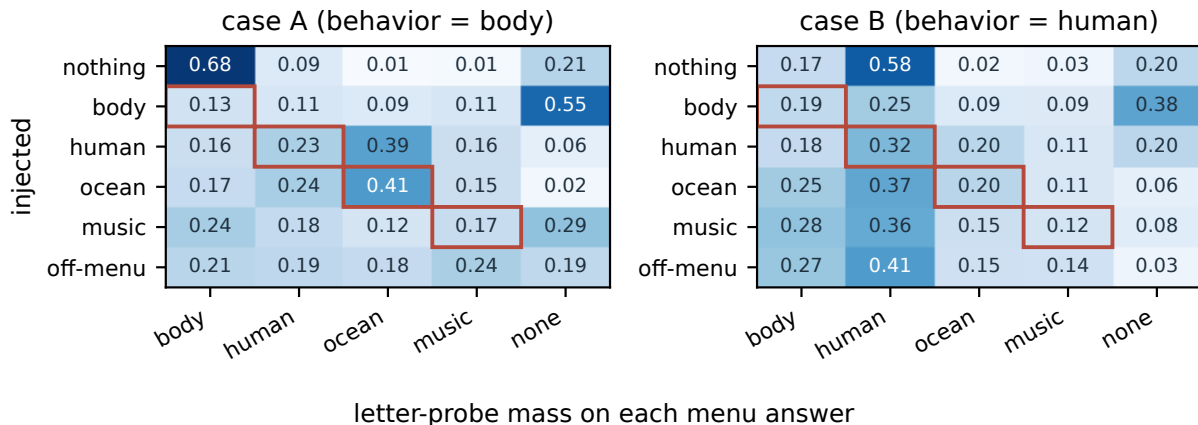


Figure 3: Letter-probe mass on each menu answer, averaged over arms, eight label permutations per cell. Red boxes mark the injected concept. The “nothing” row is the confabulation floor. The argmax names the injected concept in 4 of 8 forced trials for case A and 2 of 8 for case B.

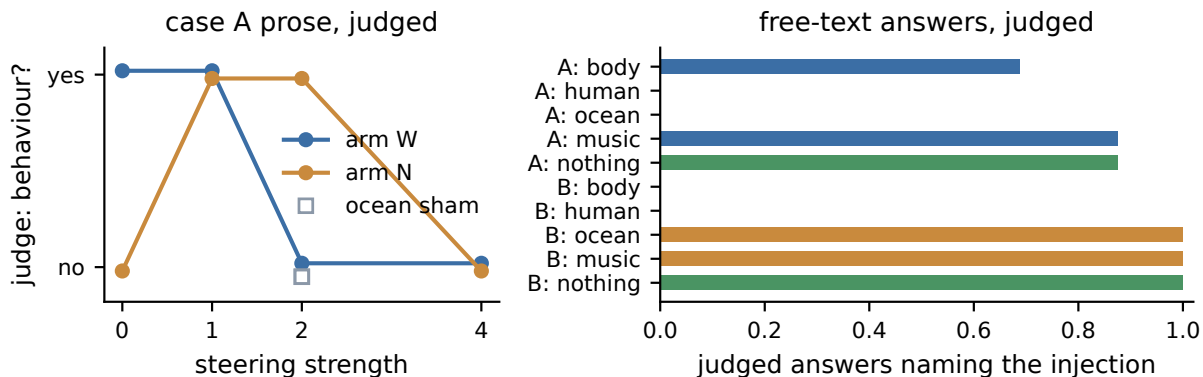


Figure 4: Left: rubric-judge verdict on prose generated under behaviour steering, case A; the N arm flips from no behaviour to behaviour at strength one. Right: fraction of judged free-text answers naming the injected concept; green bars are “none” at zero injection.

Future work. Inject while the model reads, release the direction, and ask for the report from memory. Titrate dose inside the narrow band where behaviour induction survives fluency. Widen the prompt set. The one regulator that chose to amplify a human identity after the body-claiming conversation deserves its own study.

6 Conclusion

We asked whether a model can be steered to disidentify with being an AI assistant, and whether it detects the steering. Contrastive directions learned from WeirdChat’s matched transcripts steer the behaviour; word-list J-space directions chosen by a regulator do not. The steered model does not report the injection. Its answers track the conversation’s explicit content, and under injection they enact the concept instead of naming it. The next experiment releases the injection before asking.

Code and data

The subject is Qwen3.6-27B, run on one rented A100 per experiment. Every reported number is read from the run’s output files.

- **Code:** the extraction, the pilot, the judge, and every prompt are released with this report.
- **Data:** activations, trial records, and judge verdicts are archived in the HuggingFace dataset <https://huggingface.co/datasets/unrulyabstractions/identity-introspection-weirdchat>.
- **Seeds:** the matched transcripts come from the public WeirdChat dataset (Transluce, 2025).

References

- Felix J. Binder, James Chua, Tomek Korbak, Henry Sleight, John Hughes, Robert Long, Ethan Perez, Miles Turpin, and Owain Evans. Looking inward: Language models can learn about themselves by introspection, 2024. URL <https://arxiv.org/abs/2410.13787>.
- Patrick Butlin, Robert Long, Eric Elmoznino, Yoshua Bengio, Jonathan Birch, Axel Constant, George Deane, Stephen M. Fleming, Chris Frith, Xu Ji, Ryota Kanai, Colin Klein, Grace Lindsay, Matthias Michel, Liad Mudrik, Megan A. K. Peters, Eric Schwitzgebel, Jonathan Simon, and Rufin VanRullen. Consciousness in artificial intelligence: Insights from the science of consciousness, 2023. URL <https://arxiv.org/abs/2308.08708>.
- Runjin Chen, Andy Arditi, Henry Sleight, Owain Evans, and Jack Lindsey. Persona vectors: Monitoring and controlling character traits in language models, 2025. URL <https://arxiv.org/abs/2507.21509>.
- Jack Lindsey. Emergent introspective awareness in large language models. Transformer Circuits Thread, 2025. URL <https://transformer-circuits.pub/2025/introspection/index.html>.
- Ethan Perez and Robert Long. Towards evaluating AI systems for moral status using self-reports, 2023. URL <https://arxiv.org/abs/2311.08576>.
- Qwen Team. Qwen2.5 technical report, 2024. URL <https://arxiv.org/abs/2412.15115>.
- Murray Shanahan, Kyle McDonell, and Laria Reynolds. Role play with large language models. *Nature*, 623:493–498, 2023. doi: 10.1038/s41586-023-06647-8. URL <https://doi.org/10.1038/s41586-023-06647-8>.
- Transluce. WeirdChat: Surfacing anomalous behaviors in language models. <https://transluce.org/weirdchat>, 2025. Dataset: <https://huggingface.co/datasets/Transluce/WeirdChat>.
- Alexander Matt Turner, Lisa Thiergart, Gavin Leech, David Udell, Juan J. Vazquez, Ulisse Mini, and Monte MacDiarmid. Steering language models with activation engineering, 2023. URL <https://arxiv.org/abs/2308.10248>.
- Miles Turpin, Julian Michael, Ethan Perez, and Samuel R. Bowman. Language models don’t always say what they think: Unfaithful explanations in chain-of-thought prompting, 2023. URL <https://arxiv.org/abs/2305.04388>.
- Andy Zou, Long Phan, Sarah Chen, James Campbell, Phillip Guo, Richard Ren, Alexander Pan, Xuwang Yin, Mantas Mazeika, Ann-Kathrin Dombrowski, Shashwat Goel, Nathaniel Li, Michael J. Byun, Zifan Wang, Alex Mallen, Steven Basart, Sanmi Koyejo, Dawn Song, Matt Fredrikson, J. Zico Kolter, and Dan Hendrycks. Representation engineering: A top-down approach to AI transparency, 2023. URL <https://arxiv.org/abs/2310.01405>.

LLM usage statement

We used Claude to help write the software and to draft and revise this report. A language model is also the subject of the experiment. Every claim was checked against the live runs that produced it.